Star Height for Symbolic Data Languages

Yoav Danieli

1 Atoms and the symbolic alphabet

Definition 1 (Atoms and data words)**.** *Let $\mathbb{A}$ be a countably infinite set of atoms. A finite data word is an element of $\mathbb{A}^*$.*

For every $i \geq 1$, let x_i output the current value of register i, and let ρ_i reassign register i. The symbol x_0 outputs an atom different from all currently stored register values. Define

$$\Gamma_\infty = \{x_0\} \cup \{x_i, \rho_i : i \geq 1\}.$$

For a finite register set $I \subseteq \mathbb{N}_{>0}$, let

$$\Gamma_I = \{x_0\} \cup \{x_i, \rho_i : i \in I\}.$$

2 Scopes and instances

Let $w = s_1 \cdots s_m \in \Gamma_I^*$. For each $i \in I$, the occurrences of ρ_i divide w into scopes of register i: the initial scope is before the first ρ_i, and every later scope starts immediately after a ρ_i and ends immediately before the next ρ_i, or at the end of the word. Empty scopes are allowed.

Definition 2 (Legal assignment trace)**.** *A legal assignment trace for w consists of valuations*

$$\nu_0, \nu_1, \ldots, \nu_m : I \to \mathbb{A}$$

and, for every position p with $s_p = x_0$, an atom $b_p \in \mathbb{A}$, subject to the following conditions.

(i) *ν_0 is injective.*

(ii) *If $s_p = \rho_i$, then*

$$\nu_p(j) = \nu_{p-1}(j) \quad (j \neq i)$$

and

$$\nu_p(i) \notin \{\nu_{p-1}(j) : j \in I\}.$$

Thus the new value differs from the previous value of register i and from all current values of the other registers.

(iii) *If $s_p = x_i$, then $\nu_p = \nu_{p-1}$.*

(iv) *If $s_p = x_0$, then $\nu_p = \nu_{p-1}$ and*

$$b_p \notin \nu_{p-1}(I).$$

Different x_0 occurrences are chosen independently. They may equal one another or values from inactive register scopes, provided they differ from all register values current at that position.

Definition 3 (Instance)**.** *A legal assignment trace produces a data word by scanning w from left to right and applying*

symbol	*output*
x_i	$\nu_{p-1}(i)$
x_0	b_p
ρ_i	ε.

The set of all resulting data words is denoted $\mathrm{Inst}_I(w)$*. For $K \subseteq \Gamma_I^*$, define*

$$\mathrm{Inst}_I(K) = \bigcup_{w \in K} \mathrm{Inst}_I(w).$$

This operational definition is equivalent to assigning one atom to every scope, with adjacent scopes of the same register assigned different atoms and all simultaneously current register values pairwise different.

3 Symbolic automata and symbolic regularity

Definition 4 (Symbolic automaton)**.** *For finite I, a symbolic automaton is an ordinary finite automaton over Γ_I. If it accepts $K \subseteq \Gamma_I^*$, the data language it generates is* $\mathrm{Inst}_I(K)$*.*

Definition 5 (Symbolically regular data language)**.** *A data language $D \subseteq \mathbb{A}^*$ is symbolically regular if there are finite I and regular $K \subseteq \Gamma_I^*$ such that*

$$D = \mathrm{Inst}_I(K).$$

Such a language K is a symbolic representation of D.

4 Ordinary and generalized symbolic regular expressions

Definition 6 (Ordinary expressions)**.** *Ordinary symbolic regular expressions are generated by*

$$E ::= \emptyset \mid \varepsilon \mid s \mid E \cup E \mid E \cdot E \mid E^*, \qquad s \in \Gamma_\infty,$$

with standard semantics $L_\mathrm{s}(E) \subseteq \Gamma_\infty^*$.

Definition 7 (Support)**.** *The register support of E is*

$$\mathrm{supp}(E) = \{i \geq 1 : x_i \textit{ or } \rho_i \textit{ occurs in } E\}.$$

The symbol x_0 contributes no register index.

Definition 8 (Domain-dependent complement)**.** *Generalized symbolic regular expressions additionally allow $\neg E$, with*

$$L_\mathrm{s}(\neg E) = \Gamma_{\mathrm{supp}(E)}^* \setminus L_\mathrm{s}(E).$$

Thus, if E mentions exactly x_1, x_3, ρ_1, ρ_3, then

$$L_\mathrm{s}(\neg E) = \{x_0, x_1, x_3, \rho_1, \rho_3\}^* \setminus L_\mathrm{s}(E).$$

Complement is local: in $(\neg E)F$, the universe used for $\neg E$ depends only on $\mathrm{supp}(E)$*, not on registers appearing only in F.*

5 Star height

Definition 9 (Expression height)**.** *For ordinary expressions,*

$$\begin{aligned} h(\emptyset) = h(\varepsilon) = h(s) &= 0, \\ h(E \cup F) = h(EF) &= \max\{h(E), h(F)\}, \\ h(E^*) &= 1 + h(E). \end{aligned}$$

For generalized expressions, additionally $h(\neg E) = h(E)$*.*

Definition 10 (Height of a symbolic language)**.** *For regular* $K \subseteq \Gamma_I^*$*, define*

$$h_{\mathrm{sym}}(K) = \min\{h(E) : E \textit{ ordinary and } L_{\mathrm{s}}(E) = K\},$$

and

$$h_{\mathrm{gsym}}(K) = \min\{h(E) : E \textit{ generalized and } L_{\mathrm{s}}(E) = K\}.$$

Definition 11 (Data star height)**.** *For a symbolically regular data language* D*, define*

$$h_{\mathrm{data}}(D) = \min \left\{ h_{\mathrm{sym}}(K) : \begin{array}{l} I \subseteq \mathbb{N}_{>0} \textit{ finite}, \\ K \subseteq \Gamma_I^* \textit{ regular}, \\ \mathrm{Inst}_I(K) = D \end{array} \right\}.$$

Definition 12 (Data generalized star height)**.** *For a symbolically regular data language* D*, define*

$$h_{\mathrm{gdata}}(D) = \min \left\{ h_{\mathrm{gsym}}(K) : \begin{array}{l} I \subseteq \mathbb{N}_{>0} \textit{ finite}, \\ K \subseteq \Gamma_I^* \textit{ regular}, \\ \mathrm{Inst}_I(K) = D \end{array} \right\}.$$

Both minima range over all symbolic representations, including representations using different register sets.

6 Research Problems

6.1 Easy problems

Definition 13 (Generalized star-free symbolic expression)**.** *A generalized symbolic regular expression* F *is generalized star-free if*

$$h(F) = 0.$$

Conjecture 1 (Decidability conjecture)**.** *There is an algorithm which, given an ordinary symbolic regular expression* E*, decides whether the data language generated by* E *has a generalized star-free symbolic representation.*

6.2 Moderate problems